Dheeraj Shrivastav

Software Engineer | AI Agents, Backend Systems, Web3

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SUMMARY

Software engineer building AI agents and backend services in production. At work: patient-facing WhatsApp and voice AI agents with doctor-in-the-loop review. Founder of DappDrop, an on-chain rewards protocol I architect and ship with AI coding agents, gated by 260+ Foundry tests including invariant fuzzing. TypeScript/Node.js, PostgreSQL, Solidity.

EXPERIENCE

Product Developer, Seragen Biotherapeutics — Jun 2026 - Present

Claude, Gemini, Node.js, TypeScript, WebSockets, WhatsApp Business API, Azure

- Built a WhatsApp patient assistant with an integrated AI voice agent that runs patient self-assessments end to end, reducing manual physician intervention by 65%.
- Designed a doctor-in-the-loop flow: AI drafts personalized patient reports and doctors approve them before delivery.
- Consolidated the team's separate tools into a single automated pipeline, removing manual hand-offs between steps.
- Built internal AI agents that give staff on-demand recommendations, cutting routine escalations to seniors.

Software Engineer Intern, Scalefull Technologies — Feb 2026 - May 2026

- Improved backend performance by ~25% with caching layers and REST API refactors (Node.js, TypeScript, NestJS).
- Designed and migrated PostgreSQL schemas to speed up queries on high-concurrency read paths.
- Raised test coverage on core microservices to ~90% and standardized API documentation with Swagger/OpenAPI.

Freelance Software Developer, Self-employed — Oct 2023 - Sep 2025

- Built and deployed 10+ web applications for clients (React, Next.js, Node.js, PostgreSQL), actively used by 500+ users.
- Roughly halved API response times with Redis caching and query optimization in Express/Prisma services.
- Set up Docker-based CI/CD with GitHub Actions, replacing manual deploys.

PROJECTS

DappDrop: Sybil-resistant on-chain rewards protocol (founder) — Jul 2025 - Present

Solidity, Foundry, EIP-712, Merkle proofs, Next.js, Prisma, PostgreSQL, The Graph. Pre-launch, on Sepolia testnet.

- Built with AI coding agents behind a verification gate I set up: invariant fuzzing, Slither/Aderyn, my review before deploy.
- Designed the protocol and security model: escrowed rewards, Merkle settlement with a 24h dispute window, fallback settler.
- Kept the core contract under the 24 KB EIP-170 limit by moving reward, NFT and fee logic into satellite modules.
- 240 unit + 26 invariant tests; an invariant caught a high-severity cross-campaign fund drain on the dev branch, pre-launch.
- Off-chain: EIP-712 attestation signer, SIWE auth, The Graph subgraph with RPC fallback, keeper worker, gasless relayer (built).

AWS Deploy AI: natural-language AWS deployments (solo) — 2025

TypeScript, MCP, AWS Bedrock (Claude), S3, CloudFront, Lambda, GitHub OAuth, Next.js

- MCP server + web app that analyzes a GitHub repo with an LLM and deploys it to AWS from a plain-English request.

OPEN SOURCE

jam-nodes (wespreadjam), external workflow-automation framework, 213 GitHub stars — Mar 2026

- 5 merged: Dev.to integration (27 tests), WordPress integration, webhook trigger, sliding-window rate limiter, DataForSEO ops.

SKILLS

Languages: TypeScript, JavaScript, Python, Solidity, SQL

AI: Claude, Gemini, AWS Bedrock, Groq, AI and voice agents, MCP, WhatsApp Business API

Backend: Node.js, NestJS, Express, Bun, REST, Socket.io, Prisma, PostgreSQL, Redis, OAuth, SIWE

Web3: Foundry (unit, fuzz, invariant), Slither, Aderyn, ethers.js, viem, EIP-712, Merkle proofs, The Graph

Frontend: React, Next.js, Tailwind CSS, Phaser

Infra: Docker, GitHub Actions, AWS (S3, CloudFront, Lambda, Bedrock), Linux, Vercel

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML), DIEMS, Maharashtra — 2023 - 2026

Diploma, Computer Science & Engineering, MGM Polytechnic, Maharashtra — 2020 - 2023